

Roll no. 24920001

Name of the Course: BCA
Semester: VI
Name of the Paper: Introduction to ML
Paper Code: TBC 601
Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1.

- a) Define Mean, Median, and Mode. Explain their significance in machine learning and Calculate Mean, Median, and Mode for the data: 12, 15, 18, 20, 22, 22, 25, 30 (10 Marks)
(CO1)
- OR
- b) Explain Range, Absolute Deviation, and Average Deviation. Find Range and Average Deviation for: 10, 12, 15, 18, 20 (CO1)

Q2.

- a. Explain the basic concepts of Probability Theory. Define sample space, event, and conditional probability with examples. (10 Marks)
(CO1)
- OR
- b. Explain different matrix operations such as Addition, Subtraction, scalar Multiplication, each with a suitable example. (CO1)

Q3.

- a. What is Machine Learning? Explain its need and importance in modern computing systems. (10 Marks)
(CO1)
- OR
- b. Explain Supervised Learning in detail. Discuss its working principle and give suitable real-world examples. (CO1)

Q4.

- a. Explain Mutable vs Immutable data types in Python Programming with examples (10 Marks)
(CO2)
- OR
- b. Discuss file modes and explain read, write, and append operations with examples. (CO2)

Q5.

- a. Explain Built-in functions & User-defined functions in Python with suitable examples (10 Marks)
(CO2)
- OR
- b. Explain the following (a) if-elif- else loop and (b) Nested loops (CO2)